

Parallel Construction of Independent Spanning Trees on Multidimensional Tori*

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Abstract

A set of spanning trees rooted at vertex r in G is called *independent spanning trees* (IST) if for each vertex v in G , $v \neq r$, the paths from v to r in any two trees are different and vertex-disjoint. If the connectivity of G is k , the IST problem is to construct k IST rooted at each vertex. The IST problem has found applications in fault-tolerant broadcasting, but it is still open for general graph with connectivity greater than four. Obokata et al. [IEICE Trans. Fundamentals of Electronics, Communications and Computer Sciences E79-A (1996) 1894–1903] have proved that the IST problem can be solved on multidimensional tori. However, their construction algorithm forbids the possibility of parallel processing. In this paper, we shall propose a parallel algorithm that is based on the *Latin square* scheme to solve the IST problem on multidimensional tori.

Keyword: independent spanning trees, internally disjoint paths, parallel algorithm, multidimensional torus, fault-tolerant broadcasting.

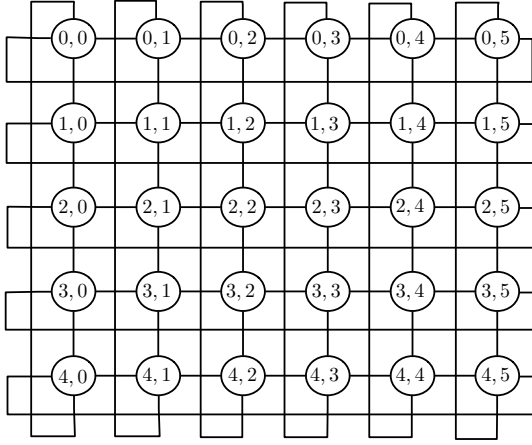
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1. Introduction

Torus has been proposed as a network topology and widely used in the design of parallel computers. In most cases, the underlying torus networks are two-dimensional. The *multidimensional torus* is a generalization of the two-dimensional torus. A d -dimensional torus, denoted by R_d or $R(m_{d-1}, m_{d-2}, \dots, m_0)$, is a graph consists of $n = \prod_{i=0}^{d-1} m_i$ vertices, where $d \geq 1$ and $m_i \geq 2$ for all i , $0 \leq i \leq d-1$. Each vertex in the torus has a unique label of the form $(a_{d-1}, a_{d-2}, \dots, a_1, a_0)$, where $0 \leq a_i \leq m_i - 1$ for all i , $0 \leq i \leq d-1$. Each vertex $(a_{d-1}, \dots, a_{i+1}, a_i, a_{i-1}, \dots, a_0)$ is connected to all of those vertices labeled by $(a_{d-1}, \dots, a_{i+1}, a_i \pm 1 \bmod m_i, a_{i-1}, \dots, a_0)$, where $0 \leq i \leq d-1$. The parameter i will be referred to as the *dimension* of the edge, while m_i is the *size* of dimension i . As for the definition of a multidimensional torus, see [1–3, 7]. For example, $R(5, 6)$ is shown in Figure 1. The sizes of dimension 1 and 0 in $R(5, 6)$ are 5 and 6, respectively.

A subclass of d -dimensional tori is the class of k -ary d -cubes. In a d -dimensional torus, different dimensions may have different sizes. However, all dimensions have the same size k in a k -ary d -cube. Note that the class of hypercubes (binary d -cubes) is also a subclass of d -dimensional tori.

Figure 1: A two-dimensional torus $R(5,6)$.

Alternately, d -dimensional tori can also be defined recursively using the operations of cartesian product, i.e., $R(m_{d-1}, m_{d-2}, \dots, m_0) = C_{m_{d-1}} \times R(m_{d-2}, \dots, m_0)$ and $R(m_0) = C_{m_0}$ [18]. Note that C_{m_i} is either a cycle with m_i vertices or K_2 in case of $m_i = 2$ for $0 \leq i \leq d-1$. By the recursive definition, it turns out that the regularity and connectivity of a d -dimensional torus are in accord with d . We denote by δ the degree of vertices in $R(m_{d-1}, \dots, m_0)$. Then, $\delta = 2\delta_1 + \delta_2$, where δ_1 is the number of $m_i > 2$ and δ_2 is the number of $m_i = 2$. For example, $R(2, 3, 2, 4)$ is a 6-regular graph since $\delta = 2\delta_1 + \delta_2 = 2 \cdot 2 + 2 = 6$.

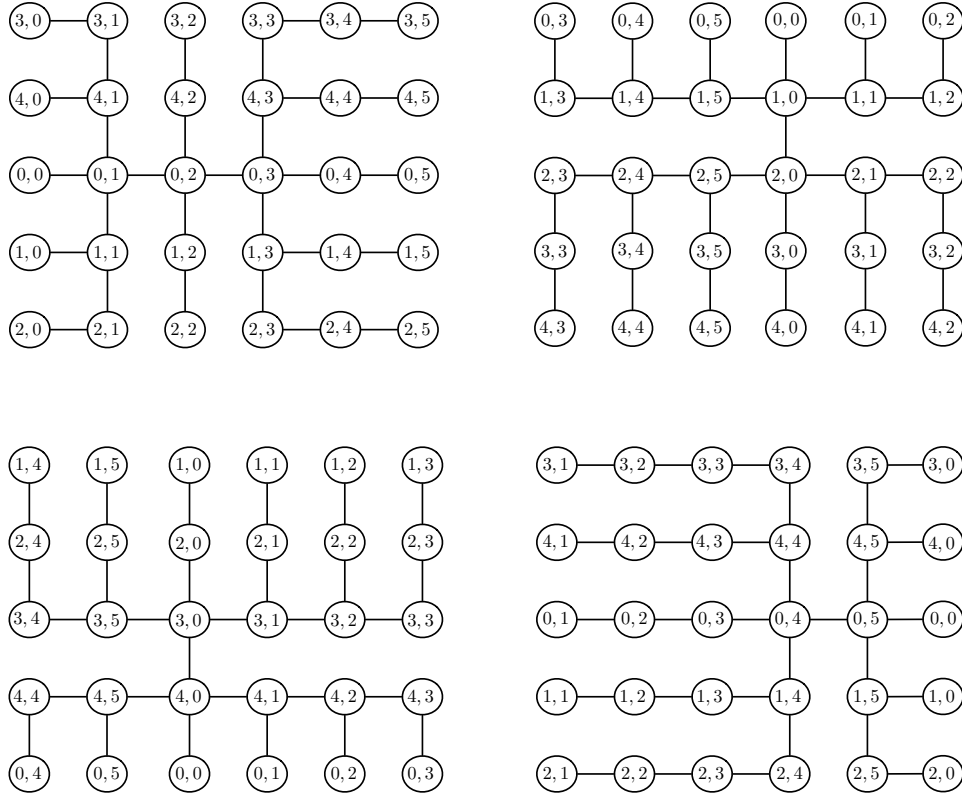
A set of different paths connecting two vertices in a graph is said to be *internally disjoint* if and only if any two paths in the set have no common vertex except the two end vertices. Considering a graph $G = (V, E)$, a *spanning tree* of G is a subgraph of G that is a tree and contains all vertices in V . Two spanning trees of G are said to be *independent* if they are rooted at the same vertex, say r , and for each vertex $v \in V \setminus \{r\}$, the two paths from r to v , one path in each tree, are internally disjoint. A set of spanning trees on a graph is said to be independent if they are pairwise independent. For example, four independent spanning trees rooted at $(0,0)$ on $R(5,6)$ are

shown in Figure 2.

The study of IST problem has contribution to the reliable communication protocols [4,13]. For example, a rooted spanning tree on a underlying graph can be viewed as a broadcasting channel for data communication. Further, the fault tolerance can be achieved by sending k copies of the message along the k ISTs rooted at the source node. If the source node is faultless, this scheme can tolerate up to $k-1$ faulty nodes and complete the broadcasting process.

Recently, the IST problem has received much attention. However, this is a very hard problem for arbitrary graphs. In fact, Zehavi and Itai [22] conjectured that for any k -connected graph G , there exist k ISTs rooted at an arbitrary vertex in G . The conjecture has been confirmed only for k -connected graphs with $k \leq 4$ in [5,6,13,22], and it is still open for arbitrary k -connected graphs when $k \geq 5$. Moreover, by providing the construction schemes, the conjecture has been proved to hold for several restricted classes of graph or digraph (especially, the graph classes related to interconnection networks), such as planar graphs [11,12,16,17], product graphs [18], chordal rings [14,20], deBruijn and Kautz digraphs [9,10], and hypercubes [19,21], etc. Note that the development of algorithms for constructing ISTs tends to pursue two research goals, one is to design efficient construction algorithms (e.g., see [14,16,17,20] for linear-time algorithms) and the other is to reduce the height of ISTs [10,19,20].

In 2004, Tang et al. first gave the definition of *optimal ISTs* and proposed an algorithm that can produce optimal ISTs for hypercubes [19]. Based on a *Hamming distance Latin square* scheme, Yang et al. proposed a parallel algorithm to generate optimal ISTs for hypercubes [21]. In this paper, we would like to generalize the parallel algorithm from hypercubes to multidimensional tori. Particularly, we will show that our algorithm relies on a simple rule and thus can easily be implemented.


 Figure 2: A set of independent spanning trees rooted at vertex $(0,0)$ on $R(5,6)$.

The remaining part of this paper is organized as follows. Section 2 introduces the underlying concept of the proposed algorithm and gives some useful properties. Section 3 presents our algorithm to generate δ ISTs on a multidimensional torus with degree δ and shows the correctness. The last section contains our concluding remarks.

2. Preliminaries

Before presenting our algorithm, we have to give some notations and properties which are necessary for illustrating our algorithm. Throughout the paper, we use the term “torus” to mean a multidimensional torus.

To explicitly represent the adjacency of vertices in a torus, we use the notation $x \xrightarrow{j} y$ to mean that vertex y is adjacent to vertex x and j is a *jump* from x to y . We use $\dim(j)$

and $\text{sign}(j)$ to express, respectively, the *dimension* and *sign* of j . Suppose x_i is the label of i -th dimension of x and m_i is the size of i -th dimension of the torus. Then, the notation $x \xrightarrow{j} y$ means that $y_{\dim(j)} = (x_{\dim(j)} + 1) \bmod m_i$ (respectively, $y_{\dim(j)} = (x_{\dim(j)} - 1) \bmod m_i$) if $\text{sign}(j)$ is positive (respectively, negative). Let J denote the set of all jumps from an arbitrary vertex in a torus. Then, the cardinality of J is the degree of the torus. If a jump $j \in J$ occurs successively to connect x and y , then we use the notation $x \xrightarrow[\alpha]{j} y$ to mean $y_{\dim(j)} = (x_{\dim(j)} + \alpha \text{sign}(j)) \bmod m_i$, where α is the number of repetitions of j . Using torus $R(3,2,4)$ as an example, we have $(2,1,3) \xrightarrow{+2} (0,1,3)$, $(2,1,3) \xrightarrow{-2} (0,1,3)$, $(2,1,3) \xrightarrow{+1(-1)} (2,0,3)$, and $J = \{+2, -2, +1(-1), +0, -0\}$.

Since a torus is vertex-symmetric, without loss of generality, we may consider the vertex $(0, \dots, 0)$ (all dimensions are 0, called it vertex 0) as the root of δ ISTs on a torus, where δ

is the degree of a vertex. Suppose that $\mathcal{T}$ is a set of δ ISTs rooted at 0 for R_d . Based on the definition of the IST problem, it is obvious that there is only one unique child for the root in every spanning tree in $\mathcal{T}$. Thus, the root can choose only one of its adjacent vertices as the child in every spanning tree. If j is the jump taken from the child to the root, we denote by $T_{\dim(j)}^{\text{sign}(j)}$ for the specified spanning tree. For example, The four ISTs of $R(5, 6)$ shown in Figure 2 are, starting from the top-left one and going clockwise, denoted by T_0^- , T_1^- , T_0^+ , and T_1^+ .

For each $T \in \mathcal{T}$, the *parent* of a non-root vertex x , denoted by $\text{parent}(T, x)$, is a vertex adjacent to x in the path from x to the root of T . The *ancestor set* of a vertex x in T , denoted by $\text{ancestor}(T, x)$, is the set of vertices in the path from $\text{parent}(T, x)$ to the root of T .

Let P be a shortest path joining a non-root vertex x and root vertex 0 in R_d . Then, P can be decomposed into a sequence of jumps. The set of distinct jumps in P is denoted by H_P . If a jump j appears more than once in P , we denote by $n(j)$ the number of occurrences of j . Further, let $\bar{j}$ be a jump with the same dimension and contrary sign of jump j . It is quite obvious that the following properties hold [1, 2]:

- (i) $u \neq v$ for any two vertices $u, v \in P$;
- (ii) $j_i \neq \bar{j}_k$ for any two jumps $j_i, j_k \in H_P$; and
- (iii) $n(j) \leq \lfloor \frac{m_{\dim(j)}}{2} \rfloor$ for every $j \in H_P$.

A *Latin square* is a square matrix with n^2 entries chosen from a set of n distinct elements such that none of the elements occurs twice within any row or column of the matrix [8]. A *shortest path Latin square* (SPLS for short) corresponding to P is a Latin square whose entries are chosen from the jump set H_P . In particular, we give the following definitions.

Definition 1. Suppose that P is a shortest path from vertex x to vertex 0 in R_d and let $H_P = \{j_0, j_1, \dots, j_{t-1}\}$ with $\dim(j_0) <$

$\dim(j_1) < \dots < \dim(j_{t-1})$. Then, the matrix defined below is called the *increasing SPLS* of P :

$$I(P) = \begin{bmatrix} j_0 & j_1 & \dots & j_{t-2} & j_{t-1} \\ j_1 & j_2 & \dots & j_{t-1} & j_0 \\ \vdots & & \dots & & \\ j_{t-2} & j_{t-1} & \dots & j_{t-4} & j_{t-3} \\ j_{t-1} & j_0 & \dots & j_{t-3} & j_{t-2} \end{bmatrix}.$$

In addition, the entry j_{i-1} in $I(P)$ is called the *predecessor* of j_i (denoted by $\text{pred}(I(P), j_i)$), while the entry j_{i+1} is called the *successor* of j_i (denoted by $\text{succ}(I(P), j_i)$). Note that the indices $i-1$ and $i+1$ is taken modulo t . (Hereafter, all arithmetics applied to the indices of the entries in SPLS are taken modulo $|H_P|$.)

Definition 2. Suppose that P is a shortest path from vertex x to vertex 0 in R_d and let $r = \sum_{j \in H_P} n(j)$. The *increasing SPLS* of P with repetitions, denoted by $I^*(P)$, is a matrix of size $t \times r$ obtained from $I(P)$ such that every entry j in a row of $I(P)$ is substituted by $n(j)$ repetitions of j in the same row of $I^*(P)$.

From the above definitions and the fact $n(j) \leq \lfloor \frac{m_{\dim(j)}}{2} \rfloor$ for every jump $j \in H_P$, $I(P)$ and $I^*(P)$ are identical in case of $d = 2$. Further, we can determine a shortest path from x to 0 according to the jumps in a row of $I^*(P)$. We define two types of paths as follows.

Definition 3. Suppose that P is a shortest path from vertex x to vertex 0 in R_d . Then, a *regular path* is a shortest path from x to 0, which is formed by taking jumps successively from a row of $I^*(P)$. Particularly, we denote $R = [P]_j$ to mean a regular path R that contains a specific jump $j \in H_P$ as its last jump. Moreover, an *extended path* of R , denoted by $(\alpha \cdot j')|R$, is the path obtained by adding α repetitions of jumps $j' \in J$ in the front of R . In addition, we give two constraints:

- (i) $\dim(j') = \dim(j)$, i.e., $j' = j$ or $j' = \bar{j}$;
- (ii) $1 \leq \alpha \leq \lfloor \frac{m_{\dim(j)}}{2} \rfloor$.

Example 1. We consider torus $R(4, 5, 2, 6, 3)$ and a shortest path P from vertex $(2, 3, 1, 0, 2)$

to vertex $(0,0,0,0,0)$. Then, we have

$$P: (2,3,1,0,2) \xrightarrow{+0} (2,3,1,0,0) \xrightarrow{-2} (2,3,0,0,0) \xrightarrow{\frac{+3}{2}} (2,0,0,0,0) \xrightarrow{\frac{-4}{2}} (0,0,0,0,0).$$

The two particular matrices defined above are given by

$$I(P) = \begin{bmatrix} +0 & -2 & +3 & -4 \\ -2 & +3 & -4 & +0 \\ +3 & -4 & +0 & -2 \\ -4 & +0 & -2 & +3 \end{bmatrix} \text{ and } I^*(P) = \begin{bmatrix} +0 & -2 & +3 & +3 & -4 & -4 \\ -2 & +3 & +3 & -4 & -4 & +0 \\ +3 & +3 & -4 & -4 & +0 & -2 \\ -4 & -4 & +0 & -2 & +3 & +3 \end{bmatrix}.$$

The four regular paths with respect to P are

$$\begin{aligned} [P]_{-4} &\equiv P, \\ [P]_{+0}: & (2,3,1,0,2) \xrightarrow{-2} (2,3,0,0,2) \xrightarrow{\frac{+3}{2}} (2,0,0,0,2) \xrightarrow{\frac{-4}{2}} (0,0,0,0,2) \xrightarrow{+0} (0,0,0,0,0), \\ [P]_{-2}: & (2,3,1,0,2) \xrightarrow{\frac{+3}{2}} (2,0,1,0,2) \xrightarrow{\frac{-4}{2}} (0,0,1,0,2) \xrightarrow{+0} (0,0,1,0,0) \xrightarrow{-2} (0,0,0,0,0), \\ [P]_{+3}: & (2,3,1,0,2) \xrightarrow{\frac{-4}{2}} (0,3,1,0,2) \xrightarrow{+0} (0,3,1,0,0) \xrightarrow{-2} (0,3,0,0,0) \xrightarrow{\frac{+3}{2}} (0,0,0,0,0). \end{aligned}$$

Besides, $(2 \cdot +3)[P]_{+3}$ and $(2 \cdot -3)[P]_{+3}$ are two extended paths of $[P]_{+3}$. That is,

$$\begin{aligned} (2 \cdot +3)[P]_{+3}: & (2,1,1,0,2) \xrightarrow{\frac{+3}{2}} (2,3,1,0,2) \xrightarrow{\frac{-4}{2}} (0,3,1,0,2) \xrightarrow{+0} (0,3,1,0,0) \xrightarrow{-2} (0,3,0,0,0) \xrightarrow{\frac{+3}{2}} (0,0,0,0,0), \\ (2 \cdot -3)[P]_{+3}: & (2,0,1,0,2) \xrightarrow{\frac{-3}{2}} (2,3,1,0,2) \xrightarrow{\frac{-4}{2}} (0,3,1,0,2) \xrightarrow{+0} (0,3,1,0,0) \xrightarrow{-2} (0,3,0,0,0) \xrightarrow{\frac{+3}{2}} (0,0,0,0,0). \end{aligned} \quad \square$$

The following propositions are prepared for proving the main result of this paper.

Proposition 2.1. *Let P be a shortest path from vertex x to vertex 0 in R_d . If R and R' are two distinct regular paths of P , then R and R' are internally disjoint.*

Proposition 2.2. *Suppose that P is a shortest path from vertex x to vertex 0 in R_d and let $R = [P]_j$ be a regular path of P . Suppose that Q is a shortest path from vertex y*

to vertex 0 in R_d and let $R' = (\alpha \cdot j^)|[Q]_{j'}$ be an extended path of Q such that $j' \notin H_P$. If $x \xrightarrow{j^*} y$, then R and R' are two internally disjoint paths connecting vertices x and 0 .*

Proposition 2.3. *Suppose that P is a shortest path from vertex x to vertex 0 in R_d and $j, j' \notin H_P$. Suppose that Q_y and Q_z are two shortest paths from vertices y and z , respectively, to vertex 0 in R_d such that $j \in H_{Q_y}$ and $j' \in H_{Q_z}$. Let $R = (\alpha \cdot j^*)|[Q_y]_j$ and $R' = (\beta \cdot j^\#)|[Q_z]_{j'}$ be extended paths of $[Q_y]_j$ and $[Q_z]_{j'}$, respectively. If $x \xrightarrow{j^*} y$ and $x \xrightarrow{j^\#} z$, then R and R' are two internally disjoint paths connecting vertices x and 0 .*

Example 2. We consider torus $R(5,6)$ (as shown in Figure 1) and a shortest path P from vertex $x=(2,5)$ to vertex 0 . Since $H_P = \{+0, -1\}$, two regular paths of P are $[P]_{+0}: (2,5) \xrightarrow{-1} (0,5) \xrightarrow{+0} (0,0)$ and $[P]_{-1}: (2,5) \xrightarrow{+0} (2,0) \xrightarrow{-1} (0,0)$. Let Q_y be a shortest path from vertex $y=(2,3)$ to vertex 0 . Then, an extended paths of Q_y is $[Q_y]_{-0}: (2,5) \xrightarrow{-0} (2,3) \xrightarrow{-1} (0,3) \xrightarrow{-0} (0,0)$ which is a path from vertex $(2,5)$ to the root. Similarly, let Q_z be a shortest path from vertex $z=(3,5)$ to vertex 0 . Then, an extended paths of Q_z is $[Q_z]_{+1}: (2,5) \xrightarrow{+1} (3,5) \xrightarrow{+0} (3,0) \xrightarrow{+1} (0,0)$ which is also a path from vertex $(2,5)$ to the root. Note that all of the four paths are internally disjoint.

3. Constructing ISTs on R_d

In this section, we solve the IST problem on R_d by constructing a set $\mathcal{T}$ of ISTs rooted at vertex 0 . The construction consists of two phases. In the first phase, we design a procedure for finding a shortest path P from every non-root vertex x to the root in R_d , and then determine the jump set H_P of P . In the second phase, a set of simple rules referred to H_P can be used to determine $\text{parent}(T, x)$ in each tree $T \in \mathcal{T}$.

For each non-root vertex $x = (x_{d-1}, x_{d-2}, \dots, x_0)$ in R_d , we perform the following procedure to generate the jump set H_P of a shortest path P from x to the root vertex:

Procedure GEN-JUMP-SET(x)

begin

$H_P = \emptyset$;

For $i = 0$ to $d - 1$ **do**

if $0 < x_i \leq \lfloor m_i/2 \rfloor$

$H_P = H_P \cup \{-i\}$;

else $(\lfloor m_i/2 \rfloor < x_i \leq d - 1)$

$H_P = H_P \cup \{+i\}$;

endif

enddo

end GEN-JUMP-SET

We have defined that $\bar{j}$ is a jump with the same dimension and contrary sign of jump j . Further, we denote by $H_{\bar{P}}$ the set of jumps $\bar{j}$ for all $j \in H_P$. Thus, there is a bijection from H_P to $H_{\bar{P}}$. To construct a $\mathcal{T}$ on $R(m_{d-1}, m_{d-2}, \dots, m_0)$, we perform the following procedure:

Procedure GEN-PARENTS(x)

begin

Step 1. Determine H_P , $H_{\bar{P}}$, and

$J \setminus (H_P \cup H_{\bar{P}})$ for x ;

Step 2. **for all** $j \in H_P$ **do**

$k = \text{pred}(I(P), j)$;

$x \xrightarrow{j} \text{parent}(T_{\dim(k)}^{\text{sign}(k)}, x)$;

enddo

Step 3. **for all** $j \in H_{\bar{P}}$ **do**

$x \xrightarrow{j} \text{parent}(T_{\dim(j)}^{\text{sign}(j)}, x)$;

enddo

Step 4. **for all** $j \in J \setminus (H_P \cup H_{\bar{P}})$ **do**

$x \xrightarrow{j} \text{parent}(T_{\dim(j)}^{\text{sign}(j)}, x)$;

enddo

end GEN-PARENTS

Example 3. We consider again torus $R(4, 5, 2, 6, 3)$ as an example. Table 1 shows the three types of jumps for $(2, 3, 1, 0, 2)$ and

its nine neighbors in the torus. According to the rules in Procedure GEN-PARENTS, information for calculating the parents of vertices $(2, 3, 1, 0, 2)$ and $(2, 3, 0, 0, 2)$ in every spanning tree are summarized in Table 2. From this table, we can see that $(2, 3, 0, 0, 2)$ is the parent of $(2, 3, 1, 0, 2)$ in T_3^+ , while $(2, 3, 1, 0, 2)$ is the parent of $(2, 3, 0, 0, 2)$ in $T_2^+(T_2^-)$.

In what follows, we show the correctness of the Procedure GEN-PARENTS.

Lemma 3.1. *Let $x = (x_{d-1}, \dots, x_1, x_0)$ be a non-root vertex in $R(m_{d-1}, \dots, m_1, m_0)$ and suppose that P is the shortest path from x to the root. If $j \in H_{\bar{P}}$, then there exists a vertex y in the torus such that $x \xrightarrow{j} y$ for some integer $\alpha \leq \lfloor \frac{m_i}{2} \rfloor$ and $j \in H_Q$, where $i = \dim(j)$ and Q is the shortest path from y to the root. In particular, $\alpha = x_i - \lfloor \frac{m_i}{2} \rfloor$ if $\text{sign}(j)$ is negative; and $\alpha = \lfloor \frac{m_i}{2} \rfloor + 1 - x_i$ if $\text{sign}(j)$ is positive.*

Proof. Suppose that $H_P = \{j_0, j_1, \dots, j_{t-1}\}$ with $\dim(j_0) < \dim(j_1) < \dots < \dim(j_{t-1})$. Since $j \in H_{\bar{P}}$, we assume that $j = \bar{j}_p$ for some p with $0 \leq p \leq t - 1$. Then, let $y = (y_d, y_{d-1}, \dots, y_0)$ be a vertex in the torus and assume that $x \xrightarrow{j} y$. We consider the following two cases:

Case 1: $\text{sign}(j_p)$ is positive. In this case, $\lfloor \frac{m_i}{2} \rfloor < x_i \leq m_i - 1$. If $\alpha = x_i - \lfloor \frac{m_i}{2} \rfloor$, then $y_i = \lfloor \frac{m_i}{2} \rfloor$. Applying procedure GEN-JUMP-SET to y , we get $j \in H_Q$.

Case 2: $\text{sign}(j_p)$ is negative. In this case, $0 < x_i \leq \lfloor \frac{m_i}{2} \rfloor$. If $\alpha = \lfloor \frac{m_i}{2} \rfloor + 1 - x_i$, then $y_i = \lfloor \frac{m_i}{2} \rfloor + 1$. Applying procedure GEN-JUMP-SET to y , we also get $j \in H_Q$. $\square$

Lemma 3.2. *Let x be a non-root vertex in $R(m_{d-1}, \dots, m_1, m_0)$ and suppose that P is the shortest path from x to the root. If $j \in J \setminus (H_P \cup H_{\bar{P}})$, then there exists a vertex y in the torus such that $x \xrightarrow{j} y$ and $\bar{j} \in H_Q$, where Q is the shortest path from y to the root.*

Table 1: Three types of jumps for vertex $(2, 3, 1, 0, 2)$ and its neighbors in $R(4, 5, 2, 6, 3)$

x	H_P	$H_{\overline{P}}$	$J \setminus (H_P \cup H_{\overline{P}})$
$(2, 3, 1, 0, 2)$	$\{+0, -2, +3, -4\}$	$\{-0, +2, -3, +4\}$	$\{+1, -1\}$
$(2, 3, 1, 0, 0)$	$\{-2, +3, -4\}$	$\{+2, -3, +4\}$	$\{+0, -0, +1, -1\}$
$(2, 3, 1, 0, 1)$	$\{-0, -2, +3, -4\}$	$\{+0, +2, -3, +4\}$	$\{+1, -1\}$
$(2, 3, 1, 1, 2)$	$\{+0, -1, -2, +3, -4\}$	$\{-0, +1, +2, -3, +4\}$	$\emptyset$
$(2, 3, 1, 5, 2)$	$\{+0, +1, -2, +3, -4\}$	$\{-0, -1, +2, -3, +4\}$	$\emptyset$
$(2, 3, 0, 0, 2)$	$\{+0, +3, -4\}$	$\{-0, -3, +4\}$	$\{+1, -1, +2, -2\}$
$(2, 4, 1, 0, 2)$	$\{+0, -2, +3, -4\}$	$\{-0, +2, -3, +4\}$	$\{+1, -1\}$
$(2, 2, 1, 0, 2)$	$\{+0, -2, -3, -4\}$	$\{-0, +2, +3, +4\}$	$\{+1, -1\}$
$(3, 3, 1, 0, 2)$	$\{+0, -2, +3, +4\}$	$\{-0, +2, -3, -4\}$	$\{+1, -1\}$
$(1, 3, 1, 0, 2)$	$\{+0, -2, +3, -4\}$	$\{-0, +2, -3, +4\}$	$\{+1, -1\}$

Table 2: The parents of vertices $(2, 3, 1, 0, 2)$ and $(2, 3, 0, 0, 2)$ in each IST of $R(4, 5, 2, 6, 3)$

x	the condition of jump j	IST	parent of x
$(2, 3, 1, 0, 2)$	$j = -0 \in H_{\overline{P}}$	T_0^-	$(2, 3, 1, 0, 1)$
	$j = +0 \in H_P$	$T_2^-(T_2^+)$	$(2, 3, 1, 0, 0)$
	$j = -1 \in J \setminus (H_P \cup H_{\overline{P}})$	T_1^+	$(2, 3, 1, 5, 2)$
	$j = +1 \in J \setminus (H_P \cup H_{\overline{P}})$	T_1^-	$(2, 3, 1, 1, 2)$
	$j = -2(+2) \in H_P$	T_3^+	$(2, 3, 0, 0, 2)$
	$j = -3 \in H_{\overline{P}}$	T_3^-	$(2, 2, 1, 0, 2)$
	$j = +3 \in H_P$	T_4^-	$(2, 4, 1, 0, 2)$
	$j = -4 \in H_P$	T_0^+	$(1, 3, 1, 0, 2)$
	$j = +4 \in H_{\overline{P}}$	T_0^+	$(3, 3, 1, 0, 2)$
$(2, 3, 0, 0, 2)$	$j = -0 \in H_{\overline{P}}$	T_0^-	$(2, 3, 0, 0, 1)$
	$j = +0 \in H_P$	T_3^+	$(2, 3, 0, 0, 0)$
	$j = -1 \in J \setminus (H_P \cup H_{\overline{P}})$	T_1^+	$(2, 3, 0, 5, 2)$
	$j = +1 \in J \setminus (H_P \cup H_{\overline{P}})$	T_1^-	$(2, 3, 0, 1, 2)$
	$j = -2(+2) \in J \setminus (H_P \cup H_{\overline{P}})$	$T_2^+(T_2^-)$	$(2, 3, 1, 0, 2)$
	$j = -3 \in H_{\overline{P}}$	T_3^-	$(2, 2, 0, 0, 2)$
	$j = +3 \in H_P$	T_4^-	$(2, 4, 0, 0, 2)$
	$j = -4 \in H_P$	T_0^+	$(1, 3, 0, 0, 2)$
	$j = +4 \in H_{\overline{P}}$	T_4^+	$(3, 3, 0, 0, 2)$

Proof. Let $i = \dim(j)$. It is obvious that $x_i = 0$ and $y_i = 1$ (or $m_i - 1$) as $\text{sign}(j)$ is positive (or negative, respectively). As a result, we get $\bar{j} \in H_Q$. $\square$

Lemma 3.3. *Procedure GEN-PARENTS can construct a set of δ spanning trees rooted at vertex 0 in R_d , where δ is the degree of R_d .*

Proof. In Procedure GEN-PARENTS, every non-root vertex x has different parents in δ different “graphs” (the cardinality of the jump

set J is δ). It turns out that every resulting graph is a spanning subgraph of R_d . To complete this proof, we need to show that there exists a unique path from every vertex x to 0 in each subgraph. Let P be a shortest path from x to 0 in R_d and let $H_P = \{j_0, j_1, \dots, j_{t-1}\}$ be a jump set with $\dim(j_0) < \dim(j_1) < \dots < \dim(j_{t-1})$. According to the type of jump j from x , we divide all subgraphs into three classes and consider the following cases:

Case 1: $j \in H_P$. Suppose $j = j_p$ for

$p = 0, 1, \dots, t-1$. We can find the unique path from x to 0 in the spanning subgraph by choosing a row in $I^*(P)$, with j_p as the first element and j_{p-1} (i.e., $\text{pred}(I(P), j_p)$) as the last element. Thus, the spanning subgraph must be a tree.

Case 2: $j \in H_{\bar{P}_x}$. Suppose x has the representation $(x_{d-1}, x_{d-2}, \dots, x_0)$ and $i = \dim(j)$. By Lemma 3.1, there exists a vertex y in the torus, such that $x \xrightarrow{\frac{j}{\alpha}} y$ and $j \in H_Q$ (Q is the shortest path from y to 0), where

$$\alpha = \begin{cases} x_i - \lfloor \frac{m_i}{2} \rfloor & \text{if } \text{sign}(j) \text{ is negative,} \\ \lfloor \frac{m_i}{2} \rfloor + 1 - x_i & \text{if } \text{sign}(j) \text{ is positive.} \end{cases}$$

As a result, we can find a path from x to 0 in the spanning subgraph by choosing a row in $I^*(Q)$, with $\text{succ}(I(Q), j)$ as the first element and j as the last element, and then extending the path from y to x by adding α more jumps j . This path forms the unique path connecting x and 0 in $T_{\dim(j)}^{\text{sign}(j)}$.

Case 3: $j \in J \setminus (H_P \cup H_{\bar{P}})$. By Lemma 3.2, there exists a vertex y such that $x \xrightarrow{j} y$ and $\bar{j} \in H_Q$ (Q is the shortest path from y to 0). Thus, the unique path from x to 0 in $T_{\dim(\bar{j})}^{\text{sign}(\bar{j})}$ consists of jump j (from x to y) and jumps chosen from a row in $I^*(Q)$ (with $\text{succ}(I(Q), \bar{j})$ as the first element and $\bar{j}$ as the last element).

Consequently, the resulting graphs of Procedure GEN-PARENTS are spanning trees of R_d . $\square$

We now show the independency of ISTs in $\mathcal{T}$.

Lemma 3.4. *For any $i, k \in J$, two spanning trees $T_{\dim(j)}^{\text{sign}(j)}$ and $T_{\dim(k)}^{\text{sign}(k)}$ generated by Procedure GEN-PARENTS are mutually independent.*

Proof. Let x be any non-root vertex in R_d . We should prove that the intersection of $\text{ancestor}(T_{\dim(j)}^{\text{sign}(j)}, x)$ and $\text{ancestor}(T_{\dim(k)}^{\text{sign}(k)}, x)$ contains only the root. Without loss of generality, we may consider the following cases:

Case 1: $j, k \in H_P$. The parent of x is assigned in Step 2 of procedure GEN-PARENTS. The assignment propagates upward to the root and forms the ancestor set of x . The vertices in the ancestor set coincide with the jumps chosen from a row in $I^*(P)$. Thus, based on Proposition 2.1, the intersection of two ancestor sets is the root.

Case 2: $j \in H_P$ and $k \notin H_P$. In this case, either $k \in H_{\bar{P}}$ or $k \in J \setminus (H_P \cup H_{\bar{P}})$, we can prove (by Proposition 2.2) that the intersection of two ancestor sets of x is the root .

Case 3: $j, k \notin H_P$. We divide this case into three subcases: (i) $j, k \in H_{\bar{P}}$; (ii) $j, k \in J \setminus (H_P \cup H_{\bar{P}})$; and (iii) $j \in H_{\bar{P}}, k \in J \setminus (H_P \cup H_{\bar{P}})$. We can prove (by Proposition 2.3) that the intersection of two ancestor sets of x is the root.

Since vertex x is arbitrarily chosen from R_d , it turns out that $T_{\dim(j)}^{\text{sign}(j)}$ and $T_{\dim(k)}^{\text{sign}(k)}$ are mutually independent. $\square$

By combining Lemmas 3.3 and 3.4, we give the following theorem.

Theorem 1. *The algorithm of constructing δ ISTs on R_d can be done in $O(\delta n)$ time, where δ is the degree of R_d and n is the number of vertices in R_d . In particular, the algorithm can be parallelized on R_d to run in $O(\delta)$ time.*

4. Concluding remarks

In this paper, we have presented a parallel algorithm for solving the IST problem on multidimensional tori. This algorithm is a generalization of the algorithm proposed in [21]. Intuitively, The height of every spanning tree is optimal. But we have not yet proved it.

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